Toward a four-dimensional Yang–Mills construction

Notebook 38: positive full-measure embeddings and local defect bounds

Research proof draft

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Abstract

We move beyond the homogeneous local integral to exact statements about finite positive Wilson measures. A specified measurable completion of the actual local coarse map admits positive linear good and remainder operators whose sum is the exact pushforward. Their iterates retain every history and the full normalization. Under literal input-domain containment and the established local Ward identities, the good measure has the actual local analytic germ on each fixed finite regulator.

Independently, a site-plane slab argument bounds prescribed large-curvature cells under the normalized Wilson measure, uniformly in the total volume. A compact-group conditional Laplace theorem controls fluctuations in a fixed box with small boundary data. A boundary-only tree gauge joins these results: prescribed separated boxes have a joint exponentially small probability of failure of explicit local gauge/minimizer windows. These are estimates on the entire bare Wilson measure, with the denominator included.

The local box windows have not been identified with the averaged constraint windows of the iterated renormalization map. The all-history decomposition has not been bounded in a multiscale complete-density norm. We record this remaining interface, a precise match to the published large-field induction, and a regularity counterexample ruling out one insufficient shortcut. The continuum existence theorem is not established.

Relation to the construction and independence of inputs

Notebook 37 establishes a positive logarithmic drift and arbitrarily long finite weak homogeneous local histories within the working proof chain and its explicitly stated upstream hypotheses. Those histories alone do not represent the complete Wilson measure. Here we address that next interface.

There are two distinct kinds of result below. The positive embedding uses the previously defined ordinary composite coarse map, stationary chart, pivot measure and local gauge Ward identity. Its local-germ assertion is conditional on literal containment of the real integration window in the represented input density. By contrast, the bare Wilson probability results are proved directly from compact-group integration, finite lattice geometry, and positivity. They do not require the earlier multiscale determinant calculation.

All field and coupling arguments in the new probability bounds are real. The compact gauge groups and link Haar measures are normalized to mass one. Unless explicitly stated otherwise, a fixed-box constant may depend on its side length and the group, but not on the surrounding torus volume. Such a constant is not claimed uniform in a growing physical box at vanishing lattice spacing. The general $SU(N)$ bare-measure lemmas do not extend all earlier $SU(2)$ construction inputs to arbitrary groups.

The source repository remains pinned at `e392ef45013a581664bab76b90867170b30fe10e`. This document is a mathematical research draft with separate internal audits, not an assertion of external verification of the preceding notebooks. The exact status of each result and each missing implication is part of its statement.

1 An exact positive finite-measure embedding of the retained sector

Fix a finite compatible periodic lattice and a compact connected matrix gauge group, with normalized link Haar measure. This section constructs an exact positive decomposition of the original finite Wilson measure under a specified real measurable coarse map. It does not estimate the complementary histories or identify the full pushforward with the homogeneous analytic density.

The primitive ordinary mean, composite ratio $R = \ell^s$, direct-R internal contour gauge, physical centered branch $V_*(W)$, and retained pivot chart are the ones already established for the local macrostep. A global completion is fixed explicitly below. It agrees with that analytic map near the identity, but is not identified with an unspecified different large-field completion in the deposited account.

1.1 The global real map and its exact compact-gauge normalization

For a contour tuple $(U_1, \dots, U_n)$ from y to x, set $T = U_1$. On a sufficiently small conjugation-invariant identity neighborhood, let Φ be the actual selected local mean on the reference-normalized tuple, with its original ordering, multiplicities and reference. On the Borel good set where all $U_j T^{-1}$ lie in that neighborhood, define

$$P(U_1, \dots, U_n) = \Phi(U_1 T^{-1}, \dots, U_n T^{-1}) T.$$

On its complement put $P = T$. The good set is endpoint-invariant, and local conjugation covariance gives, everywhere on the real group,

$$P(gU_1 h^{-1}, \dots, gU_n h^{-1}) = gP(U_1, \dots, U_n) h^{-1}.$$

Right covariance proves agreement with the original small-field mean. No alternative group mean is substituted.

For each primitive ordinary coarse bond, compute these completed internal contours and its original center string T_c. If every transported relative string $U_x T_c^{-1}$ lies in the specified small logarithm domain, use the original formula

$$\exp\left[\ell^{-d} \sum_x \log(U_x T_c^{-1})\right] T_c;$$

otherwise use T_c. This defines a Borel gauge-covariant primitive map. Compose these primitive completions to obtain F. Complete the independent direct-R internal contour P_R in the same way. Thus, with r restriction to final block centers,

$$F(U^v) = F(U)^{r(v)}, \qquad P_R(U^v; y, x) = v(y) P_R(U; y, x) v(x)^{-1}. \tag{1}$$

The completion need not be continuous at its selected boundaries and is not asserted globally holomorphic. The original fine Wilson measure is unchanged: F is now a fully specified coarse observable of it. No equality with a direct-R ordinary average is used.

Let H_R be the site-gauge group equal to the identity at every final block center, with product normalized Haar, and let n_s count its free site variables. For positive real g, define

$$q(h) = 1 - \operatorname{Re} \tau(h), \quad G_R(U) = \sum_{y, x \neq y} q(P_R(U; y, x)), \quad c(g) = \int_G e^{-q(h)/g^2}\, dh > 0. \tag{2}$$

This is the exact compact integral, not a Gaussian approximation. Equation (1) and independence of the site Haar variables give, for every real configuration,

$$\int_{H_R} e^{-G_R(U^u)/g^2}\, du = c(g)^{n_s}. \tag{3}$$

Indeed its individual contour factors are $P_R(U; y, x) u(x)^{-1}$, each with Haar integral $c(g)$.

Lemma 1.1 (Pointwise orbit acceptance)**.** *The kernel*

$$K_g(U, dV) = c(g)^{-n_s} \int_{H_R} e^{-G_R(U^u)/g^2}\, \delta_{U^u}(dV)\, du \tag{4}$$

is a probability kernel supported on $F(V) = F(U)$. For every Borel $0 \le \chi(V, W) \le 1$, put $p_\chi(U) = \int K_g(U, dV)\chi(V, F(U))$. Then for every finite positive measure μ,

$$F_*\mu = F_*(p_\chi \mu) + F_*((1 - p_\chi)\mu), \qquad 0 \le p_\chi \le 1. \tag{5}$$

If μ is H_R-invariant, then

$$K_g^*\mu(dV) = c(g)^{-n_s} e^{-G_R(V)/g^2} \mu(dV). \tag{6}$$

Proof. Normalization follows from (3); based gauge invariance of F gives the support statement. The split is therefore pointwise acceptance/rejection. For the last assertion, in the integral against each test function change U to U^u, using invariance of μ, then integrate du. For noninvariant input the corresponding formula uses its H_R-average, not its original density. □

1.2 The actual local density and its positive domination

Let μ be invariant under the full current-site gauge group and write its Haar Lebesgue decomposition $\mu = \rho\, m_E + \mu_{\text{sing}}$. Both parts are invariant by uniqueness of the decomposition and Haar invariance. Use only ρm_E for the local coarea operator; every remaining positive singular part will stay in the complement.

For W in a small real domain D, set $V = V_*(W)$. The actual chart $\Phi_V(z)$ solves the relative constraint exactly and leaves every retained coordinate equal to z. Choose an injective local sheet with a fixed real retained window $\mathcal{B}_z$. An additional factor $0 \le \chi_{\text{extra}} \le 1$ may impose any further original pivot, contour or reconstructed-field restrictions; those factors are not silently dropped.

Let n_e, n_c count positive current and coarse bonds, and set $n_z = (\dim \mathfrak{g})(n_e - n_c)$. In real Lie coordinates orthonormal for the chosen trace form, write $d(e^{iX}) = h_0 j(X) dX$, with $j(0) = 1$. For the actual pivot blocks $A_{c,p}$, define

$$D_R(V) = \prod_c \det(A_{c,p})^{-1}.$$

All real small-chart orientations are fixed positive. Let $j_c^{\text{str}}(V; K_V z)$ be the logarithm of $\det A_{c,p} / \det D_p F_V(\Phi_V(z); c)$, zero at $z = 0$. The exact local density relative to coarse product Haar is

$$\begin{aligned} \ell_\mu(W) = c(g)^{-n_s} h_0^{n_e - n_c} D_R(V) \int_{\mathcal{B}_z} \rho(e^{i\Phi_V(z)} V) e^{-G_R(e^{i\Phi_V(z)}V)/g^2} \\ \times \prod_b j(\Phi_V(z)(b)) \exp\left[\sum_c j_c^{\text{str}}(V; K_V z)\right] \chi_{\text{extra}}\, dz. \end{aligned} \tag{7}$$

The target group delta contributes h_0^{-1} per coarse bond. The square ambient straightening and subsequent pivot elimination give this expression; a rectangular $\det D\Phi$ is not substituted for the constraint Jacobian. Equivalently its unrearranged pivot factor is the product of inverse nonlinear pivot derivatives. For measurable ρ, the local joint change of variables and Tonelli prove (7) almost everywhere. It is a nonnegative measurable function. Applying (6) to this actual local sheet gives

$$\nu := F_*\mu \;\ge\; 1_D \ell_\mu\, m_C. \tag{8}$$

No global density or regularity for ν is required.

If the literal input chart has $\rho(U) = Z_{\text{in}} \exp[-S_M(U)/g^2 + E_{\text{raw}}(U)]$, put $z = g\xi$. With the actual pulled-back action-plus-gauge Hessian $A_M(V)$, its exact cubic-and-higher remainder and all measure factors in I, the same formula is

$$\begin{aligned}\ell_\mu(W) = {} & Z_{\text{in}} c(g)^{-n_s} h_0^{n_e - n_c} D_R(V) g^{n_z} (2\pi)^{n_z/2} \det A_M(V)^{-1/2} \\ & \times e^{-S_M(V)/g^2 + E_{\text{raw}}(V)} \mathbb{E}_{A_M(V)^{-1}}[\chi_{\text{window}} \chi_{\text{extra}} e^{I(V,g,\xi)}].\end{aligned} \tag{9}$$

Here I contains $-g^{-2}R_3$, all Haar and straightening logarithms, and the exact inherited fluctuation difference. Its field argument is $\Phi_V(g\xi)$, not $g\Phi_V(\xi)$. This is an exact positive-real identity only when the input density representation holds at every reconstructed field of the window. Holomorphic continuation alone does not furnish that condition.

1.3 An invariant positive good/remainder pair

Choose a fixed continuous $\theta \geq 0$ supported in D, positive near the identity. Let $\mathcal{G}_C$ be the full coarse-site gauge group with normalized Haar. Define

$$m_\theta(W) = \int_{\mathcal{G}_C} \theta(W^v)\, dv.$$

It is invariant, and positive on the orbit saturation of $\{\theta > 0\}$. With value zero when $m_\theta = 0$, set

$$a_\mu(W) = \frac{\int_{\mathcal{G}_C} \theta(W^v) \ell_\mu(W^v)\, dv}{m_\theta(W)}, \qquad \mathcal{A}_g \mu = a_\mu m_C. \tag{10}$$

Only real values of the locally computed density are used, where θ is supported. This is not a globally analytic extension.

Theorem 1.2 (Exact invariant positive split)**.** *On the cone of full-gauge-invariant finite positive measures,*

$$0 \leq \mathcal{A}_g \mu \leq F_* \mu, \qquad \mathcal{R}_g \mu := F_* \mu - \mathcal{A}_g \mu \geq 0. \tag{11}$$

Both operators are linear, preserve gauge invariance, and satisfy $\mathcal{A}_g + \mathcal{R}_g = F_*$, including raw mass.

Proof. The full pushforward is invariant by (1). Its Haar-absolutely-continuous part has an invariant nonnegative density f, chosen by compact gauge averaging. Equation (8) implies $f \geq \ell_\mu$ almost everywhere on D. For almost every W, v,

$$\theta(W^v)\ell_\mu(W^v) \leq \theta(W^v) f(W^v) = \theta(W^v) f(W).$$

Integrating in v and dividing on $m_\theta > 0$ proves $a_\mu \leq f$. The positive singular part remains in the remainder. Tonelli, or truncation of the positive factor m_θ^{-1}, justifies this near its zero set. Linearity follows from the linear Haar-absolutely-continuous projection, local fiber integration and fixed θ. The remainder is positive by the proved inequality. □

The proof cannot be replaced by averaging a measure inequality and then dividing it by m_θ. It uses the invariant density bounding each orbit value before normalization. No uniform lower bound on m_θ is needed. This remainder need not be the literal rejection in (5): the normalized output average redistributes the local sector along coarse gauge orbits. Both splits are exact.

1.4 The local germ is retained on a fixed finite regulator

Lemma 1.3 (Finite orbit-overlap control)**.** *Let ℓ be an analytic local density on a product potential ball of radius r on a connected finite coarse lattice. Assume its total infinitesimal gauge Ward identity holds there and it is invariant under constant color conjugations. There is $\epsilon > 0$, possibly depending on the finite graph, such that*

$$W, W^v \in D_\epsilon \quad \Longrightarrow \quad \ell(W^v) = \ell(W). \tag{12}$$

Proof. Use a unitarily invariant matrix norm, paying the fixed color-coordinate equivalence constants in r. Choose a rooted spanning tree with maximal root-path length D_{tree}. Write $v_x = h_x c$, where $c = v_o$, $h_o = I$. Constant conjugation does not increase the invariant field norm. On each tree edge xy,

$$h_y = (W_{xy}^v)^{-1} h_x W_{xy}^c.$$

Telescoping unitary products gives $\|h_x - I\| \le C D_{\mathrm{tree}} \epsilon$. Choose ϵ so every $h_x = e^{i\lambda_x}$ uses one small logarithm chart, with $\|\lambda_x\| \le C D_{\mathrm{tree}} \epsilon$. Along $h_x(t) = e^{it\lambda_x}$, the transformed angles are bounded by $C(1 + D_{\mathrm{tree}})\epsilon$. Taking $\epsilon \le cr/(1 + D_{\mathrm{tree}})$ keeps the full real path inside the Ward domain. Integrate that identity, then apply constant conjugation invariance. No global logarithm of the original v was assumed. □

For the actual chosen sector, this application uses a product of *invariant link/color balls* and $\chi_{\mathrm{extra}} = 1$, or extra indicators separately proved covariant. It is not asserted for arbitrary scalar-coordinate boxes or arbitrary extra cutoffs. The exact real-window proof precedes continuation. For a small real coarse gauge k, physical branch uniqueness and the block-constant lift give $V_*(W^k) = V_*(W)^{k^{\mathrm{up}}}$. With R_k the source-endpoint adjoint on relative fields and D_k its retained restriction, constraint equivariance and chart uniqueness imply

$$\Phi_{V_*(W^k)}(D_k z) = R_k \Phi_{V_*(W)}(z).$$

Each retained adjoint is orthogonal and has determinant one. Hence it preserves the sharp product of color balls exactly. Input gauge invariance, conjugation invariance of the absolute contour penalty and the actual Haar/pivot change of variables prove equality of the *literal real retained integrals*. No cutoff boundary term is omitted. These are the exact identities proved in the preceding local macrostep covariance argument; the analytic density then extends the differentiated Ward identity over its connected local ball. Constant conjugation follows by the same argument. The density version needs no new logarithm branch. For other cutoffs this application is conditional, while Theorem 1.2 remains unconditional.

Choose θ supported in D_ϵ, positive in its half ball. If W is in the half ball, then $m_\theta(W) > 0$, and every contributing W^v lies in D_ϵ. Consequently

$$a_\mu(W) = \ell_\mu(W) \qquad \text{near the identity.} \tag{13}$$

Thus a real measurable global good measure has exactly the previously computed analytic germ. The graph-dependent shrink is explicit; common-ball Ward continuation alone is not invariance between arbitrary orbit endpoints whose connecting paths may leave that ball.

1.5 Full histories, sources and raw normalization

At each finite stage fix the map, gauge penalty, positive coupling, window and output cutoff. Write $\mathcal{T}_j^{\mathrm{good}} = \mathcal{A}_j$ and $\mathcal{T}_j^{\mathrm{rem}} = \mathcal{R}_j$. Starting with the original unnormalized finite Wilson measure $\mu_0 = e^{-\gamma_0 S_W} m_E$, define for every binary word h

$$\mu_h = \mathcal{T}_{n-1}^{h_{n-1}} \cdots \mathcal{T}_0^{h_0} \mu_0.$$

Every history is a positive invariant finite measure. Repeatedly using Theorem 1.2 proves

$$(F_{n-1}\circ\cdots\circ F_0)_*\mu_0=\sum_{h\in\{\mathrm{good,rem}\}^n}\mu_h,\qquad Z_W=\sum_h\mu_h(1). \tag{14}$$

The same equality holds against every bounded terminal coarse observable. It is exact and includes all mixed histories, singular parts and raw vacuum factors; it supplies no smallness estimate for the remainder.

For bounded gauge-invariant original observables, the same linear identities apply to their invariant signed numerator measures, by positive/negative decomposition. Equivalently apply them to $e^{\sum t_iO_i}\mu_0$ on real compact source intervals and differentiate, holding step maps and windows fixed. This provides exact finite numerator and denominator bookkeeping, not uniform observable estimates in a refinement limit.

To identify an all-good germ with the previously established homogeneous map, its input density must actually equal $Z_je^{-\gamma_jS_{M_j}+E_{j,\mathrm{raw}}}$, with $\gamma_j=g_j^{-2}$, on a real neighborhood D_{in}, and one must verify the literal containment

$$e^{i\Phi_{V_*(W)}(g_j\xi)}V_*(W)\in D_{\mathrm{in}}\quad\text{for every retained }\xi\text{ and the chosen }W. \tag{15}$$

Every additional indicator is retained, unless it is proved identically one there. Under this condition (9) is the actual homogeneous local map, and (13) embeds that germ as a positive full-measure submeasure. For a fixed finite number of stages, sufficiently weak couplings and small angle windows can meet these finite margins: a logarithmic threshold is compatible with $gT(g)\to 0$. This is not a volume-uniform assertion about the tuned histories constructed earlier.

With A_0 the flat retained precision, the exact field-independent factor extracted in one stage is

$$N_j=c(g_j)^{-n_s}h_0^{n_e-n_c}g_j^{n_z}(2\pi)^{n_z/2}D_R(1)\det(A_0)^{-1/2}. \tag{16}$$

The local output is $Z_jN_je^{-\gamma_jS_{M_{j+1}}+F_{j,\mathrm{raw}}}$. The raw correction contains the inherited background, actual pivot and Gaussian determinant ratios and the exact retained expectation; its value at zero is not discarded. Since $S_{M_{j+1}}(0)=0$, if

$$F_{j,\mathrm{raw}}=\beta_jS_{M_{j+1}}+E_{j+1,\mathrm{raw}}+c_j,\qquad c_j=F_{j,\mathrm{raw}}(0)-E_{j+1,\mathrm{raw}}(0),$$

then the exact bookkeeping is

$$\gamma_{j+1}=\gamma_j-\beta_j,\qquad Z_{j+1}=Z_jN_je^{c_j}. \tag{17}$$

Replacing an input representative by $E'_{\mathrm{raw}}=E_{\mathrm{raw}}+c$ requires $Z'=Ze^{-c}$. Linearity of the orbit extension preserves all these scalars in the globally defined good submeasure. No global analytic classical action is required for these local identities.

1.6 The quantitative interface remains separate

Let ν be the full terminal pushforward and ν_{good} its all-good submeasure. If the latter has positive mass, put $\epsilon_{\mathrm{bad}}=1-\nu_{\mathrm{good}}(1)/\nu(1)$. Positivity and the mixture identity give

$$\left|\frac{\nu(f)}{\nu(1)}-\frac{\nu_{\mathrm{good}}(f)}{\nu_{\mathrm{good}}(1)}\right|\le 2\|f\|_\infty\epsilon_{\mathrm{bad}}. \tag{18}$$

No useful bound on ϵ_{bad} has been proved here. In large volume, global all-good mass may be a poor criterion; a spatial mixed-history expansion can be necessary for local observables.

The remaining tasks include uniform literal chart/orbit-domain matching, interacting coercivity and normalization estimates for the actual remainders, collared mixed-field factorization with the correct conditional means and compressed precisions, and observable bounds with the same full denominator. Homogeneous Gaussian bad-block bounds do not alone prove them. The present result is an exact positive finite bridge and an explicit remainder/history operator, not a completed global RG, continuum measure, reconstruction theorem or physical mass gap.

2 A full-measure Wilson Peierls bound

We prove a rare-event estimate under the actual normalized positive Wilson measure. It is independent of the Gaussian small-field analysis. The event controlled here is large plaquette curvature; identifying it with the complement of the actual retained-coordinate window is a separate, unresolved interface.

Let $G = SU(N)$, $m_G = \dim G$, $\tau = \mathrm{Tr}\,/N$, and

$$\Lambda = \prod_{i=1}^{d} \mathbb{Z}/n_i\mathbb{Z}, \qquad n_i = 2m_i b, \quad m_i \geq 1, \quad b \geq 3.$$

There are $V = |\Lambda|$ sites, dV independent links and $\binom{d}{2}V$ elementary unoriented plaquettes. Normalized Haar measure is used on every link. Set

$$h(W) = 1 - \mathrm{Re}\,\tau(W) = \tfrac{1}{2}\|W - I\|_\tau^2, \qquad A(U) = \sum_p h(U_p), \qquad \mu_g(dU) = Z_g^{-1} e^{-A(U)/g^2} \prod_e dU_e, \tag{19}$$

where $\|X\|_\tau^2 = \tau(X^*X)$. All side lengths are at least six, avoiding short-period graph degeneracies.

Fix any translated grid of cubes of side b. Let $\mathcal{P}_j^{\mathrm{int}}$ be the plaquettes whose vertices are strictly inside cell j, and define

$$\mathcal{B}_j(\delta) = \left\{ \max_{p \in \mathcal{P}_j^{\mathrm{int}}} h(U_p) \geq \delta^2 \right\}. \tag{20}$$

Strict interior means that every coordinate of every used link endpoint lies strictly between the corresponding walls.

Theorem 2.1 (Prescribed bad cells)**.** *There are $g_0 > 0$ and $C_{G,d} < \infty$, depending only on the group and dimension, such that for $0 < g \leq g_0$ and every set S of distinct cells,*

$$\mu_g\left(\bigcap_{j \in S} \mathcal{B}_j(\delta) \right) \leq z_b(g,\delta)^{|S|}, \quad z_b(g,\delta) = \min\left\{ 1, \exp\left[-\frac{\delta^2}{g^2} + b^d \{ d m_G \log(1/g) + C_{G,d} \} \right] \right\}. \tag{21}$$

Constants are uniform in volume and in the selected cells. The source regulators $n = 2L^M$, $L \geq 3$ odd, admit the construction with $b = L^u$ for each fixed $u \geq 1$ and $M \geq u$.

2.1 Exact site-plane slabs and the normalized chessboard inequality

Every cell wall is an integer coordinate, or *site*, plane. Its reflection preserves the lattice. An elementary plaquette cannot have vertices on both strict sides of such a plane. It either lies in the plane or belongs wholly to one adjacent closed half-space, possibly touching the plane. Conditional on tangential links in the two opposite reflection planes of the torus, the two open half configurations have reflected product Haar measures and reflected Wilson weights. The reflection quadratic form is therefore an integral of an absolute square against a nonnegative boundary weight. This proves the needed site-plane reflection positivity directly.

To obtain the chessboard inequality for an arbitrary even number of cells, cut first in one coordinate into $p = 2m$ slabs of width b. Let $\mathcal{H}$ be L^2 of the product Haar measure on tangential links in one boundary plane. Integer translation identifies successive boundary spaces.

For a bounded slab observable f, supported strictly inside the slab in the cut direction, form the integral kernel K_f by integrating all nonboundary slab links. Give it the full action of every plaquette assigned to the slab, including plaquettes touching a boundary but not lying in it, and half the action of each plaquette lying entirely in either boundary. Multiply by f. The compact integration domain and bounded integrand make K_f Hilbert–Schmidt.

Gluing the kernels by boundary Haar integration gives exactly

$$\mathrm{Tr}(K_{f_0} \cdots K_{f_{p-1}}) = Z_g \left[\prod_{j=0}^{p-1} f_j \right]. \tag{22}$$

Every boundary link is integrated once; every boundary plaquette gets two halves; every other link and plaquette is assigned once. No gauge-volume factor occurs. Reflection exchanges the two boundaries, preserves $h(U_p)$, and inverts a chosen link coordinate when its orientation reverses. Haar invariance gives

$$K_f^* = K_{\theta \overline{f}}. \tag{23}$$

For real indicators complex conjugation has no effect.

Schatten Hölder, with the even integer p, gives

$$\left| \mathrm{Tr} \prod_{j=0}^{p-1} K_{f_j} \right| \le \prod_{j=0}^{p-1} \left[\mathrm{Tr}\big((K_{f_j}^* K_{f_j})^{p/2}\big) \right]^{1/p}. \tag{24}$$

Each factor is the exact Wilson partition integral with p alternating reflected copies of one slab. The torus period remains pb. This explains precisely why the even cell count is required.

Apply (24) successively in all coordinate directions. A slab source is the product of its constituent cell sources. Orthogonal affine coordinate reflections commute on the unoriented graph, with the corresponding orientation inversions on link variables. The final factor associated with an original cell is therefore its full reflected dissemination. Intersections of cut planes require no extra normalization: every step is the exact one-axis gluing (22).

Write $N_c = V/b^d$ for the number of cells. Let Θ_j map the reference cell to cell j by the alternating reflection pattern; even counts make this pattern periodic. For a reference-cell observable f, write $\mathcal{D}(f) = \prod_i \Theta_i f$. Pull each original observable back to this reference cell. The result of the iteration is

$$\left| Z_g \left[\prod_j \Theta_j f_j \right] \right| \le \prod_j Z_g[\mathcal{D}(f_j)]^{1/N_c}. \tag{25}$$

For nonnegative indicators, all terms are nonnegative. Dividing by the *full* partition function, and using $f_j = 1$ off a selected set, gives

$$\mu_g \left(\prod_{j \in S} \Theta_j f_j \right) \le \prod_{j \in S} \left(\frac{Z_g[\mathcal{D}(f_j)]}{Z_g} \right)^{1/N_c}. \tag{26}$$

In particular, the normalization is not an independently chosen small-field partition function.

The geometry used here is simpler than a diagonal fold: all walls are site planes and no alternating site/link scheme is needed. For $n = 2L^M$, $b = L^u$, the number of cells in every direction is $2L^{M-u}$, which is even despite L being odd.

2.2 A lower bound for the full normalization

There are $\epsilon_0 > 0$ and $c_G \in (0, 1]$ such that

$$\mathrm{Haar}\{U : \|U - I\|_\tau \le \epsilon\} \ge c_G \epsilon^{m_G}, \qquad 0 < \epsilon \le \epsilon_0. \tag{27}$$

Indeed the Haar density in a fixed local exponential chart is smooth and strictly positive. This is a finite-dimensional group estimate.

If all links lie in that ϵ-ball, a four-term product telescoping and unitarity show

$$\|U_p - I\|_\tau \le 4\epsilon, \qquad h(U_p) \le 8\epsilon^2.$$

For $g \le g_0 := \min(\epsilon_0, 1)$, restrict the full integral to $\|U_e - I\|_\tau \le g$ on every link. It follows that

$$Z_g \ge (c_G g^{m_G})^{dV} \exp\left[-8\binom{d}{2}V\right], \qquad -\frac{1}{V}\log Z_g \le dm_G \log(1/g) + C_{G,d}, \tag{28}$$

where one can take

$$C_{G,d} = 8\binom{d}{2} + d\log(1/c_G).$$

This is only a lower bound for the normalization. It does not claim that the small-link box has large probability.

On the full dissemination of $\mathcal{B}_j(\delta)$, every cell contains a distinct interior plaquette of energy at least δ^2. Thus $A(U) \ge N_c\delta^2$ there and

$$Z_g[\mathcal{D}(\mathbf{1}_{\mathcal{B}})] \le e^{-N_c\delta^2/g^2}. \tag{29}$$

The remaining product Haar integral is at most one. Combining (28) and (29) bounds each factor in (26) by the exponential in (21); it is also at most one. This proves Theorem 2.1.

More generally, a cell event forcing its strictly interior action to be at least e_j has the factor

$$\min\left\{1, \exp\left[-e_j/g^2 + b^d\{dm_G \log(1/g) + C_{G,d}\}\right]\right\}. \tag{30}$$

A threshold $\|U_p - I\|_\tau \ge \delta$ uses $e_j = \delta^2/2$. At fixed b, the threshold $\delta(g) = gR(g)$ is rare whenever $R(g)^2/\log(1/g) \to \infty$, for example $R(g) = (\log(1/g))^a$ with $a > 1/2$. The explicit b^d cost precludes claiming the same constants when the cell size grows with the cutoff.

2.3 Prescribed plaquettes and Peierls clusters

Let S be any prescribed set of distinct elementary plaquettes. Color $(x; \mu, \nu)$, $\mu < \nu$, by $(x \bmod b, \mu, \nu)$. There are

$$Q_b = b^d\binom{d}{2} \tag{31}$$

colors. Choose deterministically a color occurring at least $|S|/Q_b$ times, with anchor residue a, and shift the cell grid by $a - (1, \dots, 1)$. Every selected anchor has relative coordinates $(1, \dots, 1)$; its plaquette vertices have coordinates 1 or 2 and hence are strictly interior for $b \ge 3$. Distinct anchors of the fixed color lie in distinct cells. The fixed orientation prevents two selected plaquettes at one anchor. The all-bad event on S implies the bad-cell event on this subset. Therefore

$$\mu_g\{h(U_p) \ge \delta^2 \text{ for all } p \in S\} \le z_b(g, \delta)^{|S|/Q_b}. \tag{32}$$

This is a finite deterministic coloring argument, not an independence assumption. The chessboard proof holds for each of the translated grids.

For the nearest-neighbor cell graph, of degree at most $2d$, the number of connected sets of size m containing a fixed cell is at most $(2d)^{2m}$: choose a canonical spanning tree and its depth-first traversal. The traversal has length $2(m-1)$ and determines the visited set. A cluster of size at least m contains such a connected subset of size exactly m. Consequently

$$\mu_g\{|\text{bad cluster at } j| \geq m\} \leq [(2d)^2 z_b(g,\delta)]^m. \tag{33}$$

For any fixed finite-degree plaquette adjacency graph the same argument and (32) apply. For adjacency by shared vertices, $\Delta_d = 16\binom{d}{2}$ is a safe degree bound, giving

$$\mu_g\{|\text{bad plaquette cluster at } p| \geq m\} \leq [\Delta_d^2 z_b(g,\delta)^{1/Q_b}]^m. \tag{34}$$

Periodic identifications do not increase these counting bounds. These are intersection and cluster estimates, not a claim of full stochastic domination by a product measure.

The union estimate for any bad cell is only $N_c z_b$. Thus the theorem does not assert that an arbitrarily large torus lies entirely in one small-field sector with uniformly high probability.

2.4 The remaining large-field interface

The distinction between curvature and retained-coordinate windows is essential. For a link with distinct endpoints, gauge invariance at one endpoint makes its marginal left-invariant under G. Hence

$$\mu_g\{U_e \in A\} = \mathrm{Haar}(A) \tag{35}$$

for every measurable A, independently of g. An unfixed link is therefore outside a shrinking identity window with probability tending to one. Pure-gauge fields can have all plaquettes equal to I and individual links far from I.

The actual retained fluctuation ξ instead depends on the nonlinear constraint, its minimizing background, the contour gauge and the pivot chart. No result above proves that $\|\xi\| > R$ forces the interior Wilson action to exceed cg^2R^2. Such a deterministic, quantitative implication for the exact chart would be an additional input. A local tree-gauge estimate from small plaquettes does not by itself identify its potential with ξ, or establish the required $C^{1,\alpha}$ background domain.

Global holonomy is a second obstruction. For example, $U_1(x) = e^{i\theta H/n_1}$, $U_\mu(x) = I$ for $\mu \neq 1$, has every plaquette equal to I, while its noncontractible Wilson line is $e^{i\theta H}$. Plaquette rarity cannot simply discard these sectors.

Finally, the proof uses the positive bare Wilson measure. It has not been extended here to complex source measures, arbitrary conditioned boundaries or effective actions after many RG steps. It supplies a genuine full-measure large-field input: uniform prescribed-event and cluster bounds on the original regulator. It does not yet supply the exact retained-window implication, a convergent sum over all large-field components with analytic source and locality bounds, or a complete Wilson RG history.

For context, the general reflection/chessboard framework is Fröhlich–Israel–Lieb–Simon, *Communications in Mathematical Physics* **62** (1978), 1–34, DOI 10.1007/BF01940327; gauge positivity is treated by Osterwalder–Seiler, *Annals of Physics* **110** (1978), 440–471, DOI 10.1016/0003-4916(78)90039-8. The site-plane gluing and normalization used here were proved explicitly above rather than imported with an unspecified boundary convention.

3 A normalized conditional Wilson tail in a fixed box

This section controls the entire compact Wilson integral for a fixed box with small boundary data. The constants depend on the box. It does not identify this box gauge with the nonlinear constrained coordinates of the homogeneous renormalization step.

Fix a cubical lattice box $\Omega = \{0, \dots, R\}^d$, $d = 4$, $R \geq 2$, with its full cubical cell structure. Let E_∂ be the edges in its topological boundary and $E_\circ$ the remaining edges. The boundary edge graph is connected. Fix $G = SU(N)$, with faithful normalized trace $\tau = \operatorname{Tr}/N$, and put

$$h(U) = 1 - \operatorname{Re}\tau(U) = \tfrac{1}{2}\,\|U - I\|_\tau^2\,, \qquad S_V(U) = \sum_{p \subset \Omega} h(U_p).$$

Boundary edge variables $V \in G^{E_\partial}$ are prescribed. All remaining edges have normalized Haar measure. Gauge transformations are fixed to the identity at every boundary vertex.

Choose a spanning forest $\mathcal{F}$ of the full vertex set with exactly one boundary vertex in every component, and all interior vertices included. Its edges lie in $E_\circ$. Such a forest exists: start with a spanning tree of the connected boundary graph, extend it to a spanning tree of the whole graph, and remove its boundary tree edges. There is a unique allowed gauge transformation making every forest edge the identity. Write $y \in \mathcal{Y} = G^{E_\circ \setminus \mathcal{F}}$ for the remaining edge variables in that gauge. The number of real coordinates is

$$n = (\dim G)\,|E_\circ \setminus \mathcal{F}|.$$

For this box $n > 0$. Fix an orthonormal real Lie algebra basis and write $y = \exp(ix)$ componentwise on a neighborhood of the identity. Here $\|x\|_\infty = \max_e \|x_e\|_\tau$ uses the adjoint-invariant Lie algebra norm on each edge.

Theorem 3.1 (Full conditional tail at fixed geometry)**.** *There are a neighborhood $\mathcal{V}$ of the identity boundary configuration, constants $g_0, r, c, C > 0$, and a real analytic function $x_*(V)$ on $\mathcal{V}$, with $x_*(I) = 0$, such that the following holds. The forest-gauge action has the unique global minimum $y_*(V) = \exp(ix_*(V))$. For $0 < g \leq g_0$ and $1 \leq T \leq r/g$, its normalized Wilson measure satisfies*

$$\frac{\displaystyle\int_{\mathcal{Y} \setminus W_{V,g,T}} e^{-S_V(y)/g^2}\,dy}{\displaystyle\int_{\mathcal{Y}} e^{-S_V(y)/g^2}\,dy} \leq Ce^{-cT^2}, \qquad W_{V,g,T} = \{\exp(ix) : \|x - x_*(V)\|_\infty \leq gT\}. \tag{36}$$

The displayed window is contained in a single fixed exponential coordinate neighborhood. The same ratio is the probability of the corresponding gauge-fixed event under the original, ungauged box Wilson measure.

Proof. We give the global normalization argument, including the gauge and topological points.

Exact forest gauge. Orient each forest component away from its boundary root. Recursively solving the transformation equation along its edges gives the unique interior gauge values. Changing from the original forest edge variables to these gauge values is triangular; each step is a left or right translation, or inversion, on G, hence preserves normalized Haar measure. For fixed gauge values the other edge transformations preserve Haar independently. The change of variables is therefore an exact product-Haar identity. A gauge-invariant action and an event defined through the unique forest representative are independent of the interior gauge variables. Their integrals over those variables equal one. No determinant or volume factor is discarded. Boundary edges are unchanged even when $V \neq I$; this is why a forest rooted at all boundary vertices is used.

The unique zero and its Hessian. At $V = I$, zero action means every plaquette holonomy is I. The cubical box is simply connected, so a flat edge field is a pure gauge: parallel transport from a fixed vertex is independent of the path, since elementary plaquette moves generate all path homotopies. Identity boundary edges force the vertex gauge to be constant on the connected boundary; normalize that constant to I. The field thus has an allowed gauge transformation to the identity. Uniqueness of the forest representative gives $y = I$. Consequently the continuous nonnegative function S_I has exactly one zero on the compact space $\mathcal{Y}$.

For a Lie algebra edge field a, zero on boundary and forest edges, the quadratic part is

$$\tfrac{1}{2}\sum_{p\subset\Omega}\|(da)_p\|_\tau^2\,.$$

If it vanishes, the elementary cellular Poincare lemma on a box gives $a = d\phi$. Boundary connectedness lets us choose $\phi = 0$ at every boundary vertex. Vanishing on each forest edge then gives $\phi = 0$ at every vertex. Hence $a = 0$. The Hessian in the n remaining coordinates is positive definite. This is a finite dimensional assertion; no uniform lower eigenvalue as R varies is asserted.

Uniform local well and global gap. The analytic implicit-function theorem supplies a critical point $x_*(V)$ near zero for V close to I. Choose a coordinate ball about zero on which the Hessian at all such V is uniformly positive. On a smaller fixed ball about $x_*(V)$, Taylor's theorem gives constants $c_0, C_0 > 0$ with

$$c_0|x - x_*(V)|^2 \le S_V(\exp(ix)) - S_V(y_*(V)) \le C_0|x - x_*(V)|^2. \tag{37}$$

The balls can be chosen uniformly after shrinking $\mathcal{V}$. To see a global gap, first choose a still smaller ball about zero. On the compact complement its $V = I$ action has a strictly positive minimum. Uniform continuity of S_V, and $S_V(y_*(V)) \to 0$, preserve half this gap for sufficiently small boundary data. Together with strict convexity on the coordinate ball this proves uniqueness of the global minimum and yields a fixed chart region $\mathcal{U} \subset \mathcal{Y}$ and $\eta > 0$ for which

$$S_V(y) \ge S_V(y_*(V)) + \eta \quad (y \notin \mathcal{U}). \tag{38}$$

Choose $\mathcal{U}$ small enough that the lower quadratic bound in (37) holds on all of it, and large enough that a fixed ball about every $x_*(V)$ lies in it. These choices are compatible by first making the convexity ball larger than the compact-complement ball and then shrinking $\mathcal{V}$. The product Haar coordinate density on its closure is bounded above and below by positive constants.

Denominator and local numerator. Writing $S_{\min} = S_V(y_*(V))$, integrate over the coordinate ball $|x - x_*| \le g$, reducing g_0 as needed:

$$Z_V(g) \ge c_1 g^n e^{-S_{\min}/g^2}. \tag{39}$$

Within $\mathcal{U}$, the complement of the window forces $|x - x_*| \ge gT$. The lower quadratic bound and the upper Haar density bound give, for $T \ge 1$,

$$\begin{aligned}\int_{\mathcal{U}\setminus W_{V,g,T}} e^{-S_V/g^2}\,dy &\le C_1 e^{-S_{\min}/g^2}\int_{|z|\ge gT} e^{-c_0|z|^2/g^2}\,dz \\ &\le C_2 g^n e^{-S_{\min}/g^2} e^{-c_2T^2}.\end{aligned} \tag{40}$$

The elementary Gaussian tail estimate absorbs its finite dimensional polynomial factor by reducing $c_2 > 0$.

The rest of the compact group integral. The remaining numerator is at most $e^{-(S_{\min}+\eta)/g^2}$, since product Haar mass is one. After (39) its contribution is at most $C_3 g^{-n} e^{-\eta/g^2}$. Since

$$\sup_{0<g\le g_0} g^{-n} e^{-\eta/(2g^2)} < \infty,$$

and $T \le r/g$, choosing r small and $c \le \min\{c_2, \eta/(2r^2)\}$ bounds this contribution by $C_4 e^{-cT^2}$. Reduce r also to keep the entire sup-norm window inside the coordinate chart. Combining the two numerators with the denominator proves (36). □

Corollary 3.2 (Logarithmic windows)**.** *For fixed $a > 1/2$, take $T(g) = (\log(1/g))^a$. For every $m > 0$ there are $g_m, C_m > 0$ such that*

$$\mu_{V,g}(\mathcal{Y} \setminus W_{V,g,T(g)}) \le C_m g^m \quad (0 < g \le g_m,\ V \in \mathcal{V}).$$

Proof. Here $gT(g) \to 0$ and $T(g)^2/\log(1/g) \to \infty$. Apply (36). □

Conditional interpretation. Embed the box in a larger lattice without local periodic identifications. A plaquette containing an edge of $E_\circ$ lies entirely inside the box: an edge not in the boundary has all its transverse coordinates strictly inside the corresponding intervals. Thus, after the exterior and boundary edges are fixed, the part of the bare Wilson action depending on the integrated edges is exactly S_V, up to an additive constant. Such constants cancel in the normalized law. The theorem is therefore a statement about the actual conditional bare Wilson measure whenever its boundary field belongs to $\mathcal{V}$. A prescribed boundary gauge transformation bringing the data into $\mathcal{V}$ also permits the theorem after transforming the event consistently.

The precise interface left open. This result controls a fixed-box forest-coordinate window about the unconstrained conditional minimizer. It does not control an arbitrary boundary field, a box growing with the cutoff, a nonlinear coarse-average fiber, or a mixed-history effective action whose global stability has not been proved. In particular an unconditional probability bound cannot simply be conditioned on a coarse-map value with no lower bound for the corresponding fiber normalization. The powers g^n cancel here because both integrals are estimated in the same compact conditional model. Extending that cancellation to the actual constrained local map is a remaining analytic task.

4 From curvature rarity to full-measure box windows

The two preceding probability estimates can be joined without assuming that a boundary field is typically close to the identity in an unfixed gauge. The resulting windows are defined by an explicit local tree gauge and a conditional Wilson minimizer. They still differ from the nonlinear coarse-average windows in the homogeneous renormalization construction.

Lemma 4.1 (A boundary-only gauge at fixed geometry)**.** *For the fixed four-dimensional cubical box Ω, choose a spanning tree $\mathcal{T}_\partial$ of its boundary edge graph and a root vertex. There is a finite constant L_R such that every boundary field has a boundary gauge, determined only by its boundary edges, with*

$$V_e = I \quad (e \in \mathcal{T}_\partial), \qquad \max_{e \in E_\partial} \|V_e - I\|_\tau \le L_R \max_{p \subset \partial\Omega} \|U_p - I\|_\tau .$$

The estimate concerns boundary plaquettes, not plaquettes through the interior. No assertion uniform in R is made.

Proof. Fix the root gauge to I and gauge the boundary tree edges to I by recursion. Every remaining boundary edge becomes the holonomy of its fixed tree fundamental loop. The boundary cubical complex is homeomorphic to S^3; its two-skeleton has trivial fundamental group. Each of the finitely many fundamental loops is therefore a finite product of conjugates of elementary boundary plaquette loops and their inverses, as a word in the graph edge generators. This assertion follows directly from the cellular presentation of the fundamental group: the plaquette boundary words normally generate all relations in the graph fundamental group. Choose one such finite word once for each non-tree edge and let L_R be the largest number of plaquette factors used.

For unitary matrices A_j, telescoping gives $\left\|\prod_j A_j - I\right\|_\tau \le \sum_j \|A_j - I\|_\tau$. Conjugation and inversion preserve this distance. Applying this inequality to the chosen word proves the bound, regardless of the size of its conjugating holonomies. □

Choose $\varepsilon_R > 0$ so small that $\sqrt{2}L_R\varepsilon_R$ lies inside the small boundary neighborhood in Theorem 3.1. We may and do take that neighborhood to be a conjugation-invariant product ball. Call a box boundary good when

$$h(U_p) < \varepsilon_R^2 \quad \text{for every } p \subset \partial\Omega.$$

On a good boundary, apply the boundary-only gauge in Lemma 4.1; extend its vertex values into the box arbitrarily, then perform the rooted forest gauge of Section 3, keeping the transformed boundary fixed. The final representative is independent of that arbitrary extension: two extensions differ by an interior gauge, which the unique forest normalization removes. It is also the representative obtained by gauging the full spanning tree $\mathcal{T}_\partial \cup \mathcal{F}$ to I.

Define the box bad event $D_\Omega(g,T)$ as follows: the boundary is bad, or the boundary is good and the resulting interior representative lies outside $W_{V,g,T}$. All these maps are measurable. The event is gauge invariant. Indeed, an arbitrary original gauge transformation changes the full tree representative only by constant conjugation at its root. The conditional minimizer transforms by that same conjugation, by uniqueness, and the local logarithm and edge-max norm respect conjugation.

Theorem 4.2 (Prescribed local box defects in the full Wilson measure)**.** *Fix R, b, with $b \geq 3$, and consider a periodic four-dimensional lattice with side lengths divisible by $2b$. Let $\Omega_1, \dots, \Omega_m$ be embedded translates of the fixed box with pairwise disjoint vertex sets. Put*

$$Q_b = b^4 \binom{4}{2}, \qquad N_\partial = \#\{p : p \subset \partial\Omega\} = 24R^3,$$

and let $z_b(g, \varepsilon_R)$ be the bad-cell factor of Section 2. There are constants $C, c, r, g_0 > 0$, independent of the total torus size and of the chosen boxes, such that

$$\mu_g\left(\bigcap_{i=1}^m D_{\Omega_i}(g,T)\right) \leq \left[Ce^{-cT^2} + N_\partial z_b(g,\varepsilon_R)^{1/Q_b}\right]^m \quad (0 < g \leq g_0,\ 1 \leq T \leq r/g). \tag{41}$$

In particular, for every $a > 1/2$, the bracket with $T(g) = (\log(1/g))^a$ is smaller than every fixed power of g as $g \downarrow 0$.

Proof. Condition on every edge outside the union of the boxes' interior edge sets $E_{\circ,i}$. This includes all box boundary edges. Every plaquette meeting $E_{\circ,i}$ lies wholly in Ω_i. Disjointness of the vertex sets therefore ensures that no plaquette meets two integrated edge sets. The conditional law factorizes exactly into the independent conditional Wilson laws of the individual boxes; all remaining action terms are constant under these integrations.

Write B_i for a bad boundary and F_i for the interior-window failure on a good boundary. Each B_i is measurable under the conditioning. The boundary gauge is also fixed under it, and it preserves the conditional product Haar law and the action. Theorem 3.1 consequently gives

$$\mathbb{P}(F_i \mid \text{conditioned edges}) \leq a_g\, \mathbf{1}_{B_i^c}, \qquad a_g = Ce^{-cT^2}.$$

The F_i's are conditionally independent.

For any subset J of boxes, a union bound over one offending boundary plaquette in each box, followed by the prescribed-plaquette bound of Section 2, gives

$$\mu_g\left(\bigcap_{i\in J} B_i\right) \leq \left[N_\partial z_b(g,\varepsilon_R)^{1/Q_b}\right]^{|J|}. \tag{42}$$

The selected plaquettes are distinct because the boxes have disjoint vertex sets. This is a correlated full-measure estimate; independence of the boundaries is not used.

Expand $\bigcap_i (B_i \cup F_i)$ over J, the boxes assigned the boundary-bad alternative. Conditional factorization bounds the contribution for J by $a_g^{m-|J|}\mu_g(\bigcap_{i\in J} B_i)$, after dropping other good-boundary indicators. Substituting (42) and summing the binomial expression proves (41).

Finally, the explicit bad-cell exponent has fixed threshold ε_R and fixed b:

$$z_b(g,\varepsilon_R)^{1/Q_b} \leq \exp\left[-\frac{\varepsilon_R^2}{Q_b g^2} + C_{b,G}\log(1/g) + C_{b,G}\right]$$

for sufficiently small g. This and Corollary 3.2 prove the last assertion. □

What this joins, and what it does not. The theorem gives joint rarity of actual, gauge-invariant local coordinate defects under the complete bare Wilson measure. It combines a correlated boundary estimate with conditional interior normalization and does not replace the Wilson measure by a Gaussian law. For a fixed separated grid of such boxes, the usual connected-set counting yields Peierls tails with the displayed bracket.

The representative and minimizer are those of a fixed conditional box with all its boundary edges held fixed. They are not identified with the averaged, constrained minimizer and independent retained cutoffs of the existing multiscale map. No derivative regularity, complex-source estimate, growing-box constant, or post-renormalization reflection positivity follows from this theorem. Thus it is a concrete full-measure local bridge, while the multiscale mixed-history induction remains an additional theorem to prove.

5 Local gauge coordinates and the complete-density interface

The homogeneous local trajectory does not yet identify all sectors of the full Wilson integral. This section records a precise published interface, proves a finite local gauge lemma, and shows why that lemma alone does not supply a uniformly smooth physical background.

5.1 The strongest published complete-density input

Bałaban's Theorem 1 in *Commun. Math. Phys.* 122 (1989), p. 355, supplies the complete effective-density class of *Commun. Math. Phys.* 119 (1988), Section 2, provided the effective couplings of that construction remain sufficiently small. The density bounds are (0.1), pp. 355–356; its large-field exponentiation concludes at (1.98)–(1.102), pp. 390–391. The original articles are available at CMP122-II and CMP119.

The relevant CMP119 input locations are:

pages/equations	interface
243–244, (0.1)–(0.2)	full pushforward and the additional operation
246–248, (1.1)–(1.10)	partitions and collars
254–257, (2.1)–(2.18)	multiscale determining sets and histories
258, (2.19)–(2.22)	large-field operator factorization
258–262, (2.23)–(2.42)	action, analytic domains and boundary terms.

In particular (2.38), p. 261, requires a local gauge with potential and first-derivative bounds, alongside the other multiscale regularity/current conditions. It is not a conclusion from one bare plaquette threshold. The theorem concerns the specified complete-density construction, not every small analytic local map. OCR-damaged numerical constants in these source formulas are not retranscribed here.

To use that result for our exact $F_R = M_\ell^s$ local step, one must either verify its complete induction class for this operator or prove a replacement large-field theorem. In particular: the full compact-field pushforward and partition must be specified; the good-sector term must equal our actual retained-window integral; exterior and transition data must be retained; and normalized complementary-region operations must be bounded. The source's own coupling recursion must be identified or replaced by an independently justified one. Uniform weakness of our tuned numbers alone is not equality of the two recursions.

Our strict product-ball families have an invariant total. They do not automatically supply individually invariant local orbit/current extensions or uniform analytic estimates on multiscale determining sets. Nor does composition of averaging constraints identify a one-shot window with sequential primitive windows. These are concrete transfer obligations, without an assertion that the source theorem cannot be adapted.

5.2 An explicit finite cubical tree gauge

Let G be a compact matrix Lie subgroup of $U(r)$, with operator norm. Let $\Lambda_n = \{0, \dots, n\}^d$, $d \geq 2$, carry its full cubical cell structure. Oppositely oriented bonds carry inverse matrices. For every vertex x, let $p(x)$ be the path from zero which first increases coordinate one, then coordinate two, and so on. These paths form a rooted spanning tree.

Lemma 5.1 (Finite cubical gauge bound)**.** *Suppose every elementary plaquette obeys*

$$\|U(\partial p) - I\| \leq \epsilon. \tag{43}$$

Set

$$v(x) = U(p(x)), \qquad U^v(x, y) = v(x)U(x, y)v(y)^{-1}. \tag{44}$$

Every tree edge of U^v is the identity, and each positive μ-edge satisfies

$$\|U^v(x, x + e_\mu) - I\| \leq \left(\sum_{\nu > \mu} x_\nu \right) \epsilon \leq (d - 1)n\epsilon. \tag{45}$$

Choose a fixed local exponential-chart radius $0 < \delta_G \leq 1$. If $(d - 1)n\epsilon < \delta_G$, write $U_b^v = \exp(i\eta A_b)$ in its unique local logarithm chart. For physical mesh η and side $\ell = n\eta$,

$$\|A\|_\infty \leq 2(d - 1)n\epsilon/\eta = 2(d - 1)\ell\epsilon/\eta^2. \tag{46}$$

Proof. The transformed edge is the product around the loop consisting of $p(x)$, the additional μ-edge and reverse $p(x + e_\mu)$. Move the additional edge backwards past the later coordinate segments. There are exactly $\sum_{\nu>\mu} x_\nu$ elementary square exchanges. Each exchange multiplies the loop holonomy by a conjugate of a plaquette or its inverse, and the final boundary word cancels. Unitary conjugation and inversion preserve distance to I. Moreover

$$\|W_1 \cdots W_m - I\| \leq \sum_{j=1}^{m} \|W_j - I\|$$

for unitary matrices, by telescoping. This proves (45); tree edges require no exchanges. On the chosen local chart, $\|\log W\| \leq 2\|W - I\|$, as follows from the scalar eigenangle bound. Dividing by η proves (46). □

Before the logarithm, this gauge is a finite product of link matrices and inverses. With a strict chart margin it is real analytic and has a holomorphic continuation near each admitted compact configuration. No refinement-uniform complex radius or derivative bound follows just from (45). On a general simply connected cell complex the same argument costs the filling areas of the tree fundamental loops; simple connectivity alone does not bound those areas.

For a fixed finite current box the lemma gives local logarithmic coordinates from sufficiently small plaquettes. For a fixed physical box under refinement, its sufficient potential threshold is $\epsilon = O(\eta^2 b/\ell)$. It does not yet identify relative fluctuations about a prescribed constrained minimum.

5.3 A gauge-invariant obstruction to physical smoothness

Fix $0 < \alpha < 1$, a small $f > 0$, and a nonzero Cartan generator H with $\|H\| = 1$. On a fixed physical cube with decreasing mesh η, let x denote integer lattice coordinates and put

$$U_1(x) = \exp\!\big(i\eta^2 f x_2 \chi(x_1) H\big), \qquad U_\mu(x) = I \quad (\mu \neq 1), \qquad \chi(k) = \begin{cases} 0 & k \text{ even}, \\ 1 & k \text{ odd}. \end{cases} \tag{47}$$

All non-12 plaquettes are I, and

$$U(\partial p_{12}(x)) = \exp(-i\eta^2 f\chi(x_1)H). \tag{48}$$

Thus every plaquette is within $\eta^2 f$ of I.

Proposition 5.2 (No uniform $C^{1,\alpha}$ gauge)**.** *For every fixed $B < \infty$, sufficiently fine configurations (47) admit no gauge representative $U^v = \exp(i\eta A)$ on the cube satisfying*

$$\|A\|_\infty + \|\nabla_\eta A\|_\infty + [\nabla_\eta A]_\alpha \le B. \tag{49}$$

The same assertion holds on an interior subcube containing adjacent plaquettes of both parities.

Proof. For any A satisfying (49), express a $\mu\nu$-plaquette by the exact jet variables

$$a = A_\mu(x), \quad b = A_\nu(x), \quad c = \nabla_{\eta,\mu}A_\nu(x), \quad e = \nabla_{\eta,\nu}A_\mu(x).$$

Its product is

$$P_\eta = e^{i\eta a}e^{i\eta b + i\eta^2 c}e^{-i\eta a - i\eta^2 e}e^{-i\eta b}.$$

The constant term is I and the first coefficient in η vanishes. Consequently $(P_\eta - I)/\eta^2$ extends analytically to $\eta = 0$, with uniformly bounded first jet derivatives on each fixed bounded jet ball. Adjacent a, b differ by at most $B\eta$, and adjacent c, e differ by at most $B\eta^\alpha$. Hence

$$\left\|(P_\eta(x) - I)/\eta^2 - (P_\eta(x + e_\rho) - I)/\eta^2\right\| \le C(B)\eta^\alpha. \tag{50}$$

The same inequality holds for the difference of their norms. But $\eta^{-2}\|U(\partial p) - I\|$ is gauge invariant at each plaquette. For (48) its adjacent values are zero and at least $f/2$ for sufficiently small η. This contradicts (50). □

This is an obstruction to every gauge, not merely to the chosen tree gauge. The example is already in that tree gauge, where $A_1 = \eta f x_2\chi(x_1)H$; its first derivative visibly grows like $f\ell/\eta$. The stronger invariant argument above does not assert that a bounded C^1 gauge alone is impossible. It excludes the fixed physical $C^{1,\alpha}$ background ball needed here.

There is no contradiction with the complete-density method. A raw fine field is not a uniformly smooth classical background. The useful decomposition is a regular constrained background plus rough fluctuations and explicitly retained exceptional regions. A plaquette tail or a local logarithmic chart alone does not construct that decomposition.

5.4 Flat torus holonomies and transition functions

On a periodic direction of N bonds and physical circumference $\ell = N\eta$, consider

$$U_1(x) = e^{i\theta H/N}, \qquad U_\mu(x) = I \quad (\mu \ne 1).$$

Its plaquettes are all I, whereas the noncontractible loop is $e^{i\theta H}$, unchanged up to conjugacy by a periodic gauge. If some periodic gauge has $U_b^v = e^{i\eta A_b}$ with $\|A\|_\infty \le b$, telescoping the loop gives

$$\|e^{i\theta H} - I\| \le N\eta b = \ell b. \tag{51}$$

Thus zero plaquette curvature cannot force an arbitrarily small global physical potential. Local tree gauges trivialize this flat field on contractible cubes, but their transition data retain the holonomy. On overlaps the exact relation is

$$U^{(i)}(x, y) = h_{ij}(x)U^{(j)}(x, y)h_{ij}(y)^{-1}, \qquad h_{ij} = v_i v_j^{-1}. \tag{52}$$

Those data must be retained or integrated consistently. They cannot be set to identity because local plaquettes are small.

5.5 A scale-comparison fact supplied by local shooting

Our actual weak finite trajectories obey $\gamma_j = g_j^{-2} \geq \Gamma_{\mathrm{w}} > 0$ and $|\gamma_{j+1} - \gamma_j| \leq D_{\mathrm{abs}}$. Telescoping yields, for any two stages,

$$\frac{\gamma_i}{\gamma_j} \leq 1 + \frac{D_{\mathrm{abs}}}{\Gamma_{\mathrm{w}}}|i-j|, \qquad \frac{g_i}{g_j} \leq \left(1 + \frac{D_{\mathrm{abs}}}{\Gamma_{\mathrm{w}}}|i-j|\right)^{1/2}. \tag{53}$$

For the second inequality apply the first with i, j interchanged. Both reverse ratios satisfy the same bounds. No one-step monotonicity is used. This is a useful polynomial interscale comparison; it is not asserted to match every source radius inequality without checking its constants and domains.

5.6 The next normalized interface

A direct full-measure route needs an exact gauge-invariant good/bad partition and a normalized conditional representation. Its good component must be the actual stationary-plus-fluctuation integral with its actual retained window. Its complementary replacement must be summable in component size, scale age and boundary geometry, uniformly in the allowed exterior data. A fixed-box conditional Wilson estimate is a genuine first input; its constants need not survive growing components or a nonlinear average constraint. Conversely, a normalized estimate on a coarse-average fiber cannot be inferred by conditioning an unconditional tail without controlling that fiber's normalization.

The import route makes the same obligations through the published complete-density class instead of a new direct proof. Neither route has been completed by the local trajectory or by the geometric lemmas in this section. No continuum-existence conclusion is asserted here.

6 The remaining construction theorem

The progress now has a precise measure-theoretic meaning. For one specified completion of the local coarse map, the full finite positive Wilson pushforward is the sum of positive history measures. At the bare level, large-curvature cells and explicit fixed-box gauge/minimizer defects have uniform-volume joint probability bounds. Neither assertion discards a bad sector or a vacuum factor.

What remains is to make the two constructions refer to the same iterated local coordinates and to control their interaction. A useful next theorem is a *complete-density transfer theorem* for the actual chosen coarse map. It would need to provide:

1. A spatial good/bad decomposition on determining sets with collars, prescribed exterior data, and retained holonomy/transition data. Its good-component integral must equal the existing constrained stationary-plus-fluctuation integral, with its actual independent windows, Haar factors, pivot factors and vacuum normalization.

2. A normalized replacement on each complementary component. The numerator and denominator must be bounded together, uniformly in the admissible exterior field, component geometry, preceding depth and scale history. A bound sufficient for summation over components and their histories is required. Bare probability rarity or a Gaussian conditional estimate does not by itself provide this normalized replacement.

3. Preservation of a specified complete-density class under the full step, with local analytic/source domains and coupling extraction belonging to that same class. The weak local shooting from Notebook 37 can be used only after its recursion has been identified with this full-density recursion, or after a shooting theorem has been proved for the latter.

The published large-field induction discussed above is a possible route to this theorem after its hypotheses have been matched. The new direct probability estimates are another input to a proof; they do not already verify the published induction class.

Why local rarity is not a global chart theorem. For any fixed family of M boxes, the one-box defect estimate gives a union bound proportional to M. If the number of boxes diverges with the cutoff, a small one-box probability does not imply that all boxes are good. For example, on the local tuned trajectories g_{bare} has order $n^{-1/2}$, while a logarithmic window yields a displayed estimate of order $\exp[-c(\log n)^{2a}]$. This bound does not defeat a number of microscopic boxes growing exponentially in n. This is a limitation of that union-bound inference, not a lower bound on the actual defect probability. The purpose of a component expansion is to allow defects and sum them with their correct weights, rather than assume a globally good field.

After the complete-density theorem. Continuum existence still requires common estimates for a sufficient gauge-invariant observable algebra, with renormalized short-distance insertions and the exact same denominator. Those estimates must produce a common subsequential continuum hierarchy, reflection positivity inherited from one positive regulator sequence, full Euclidean covariance, and the regularity needed for reconstruction. A quantitative lower witness and identification of the resulting fields and dynamics are needed to establish a nontrivial Yang–Mills theory. The physical mass gap is a subsequent assertion. The Gaussian gaps used in the local analysis are not that spectral statement.

No uniqueness of the continuum limit is imposed merely for the sake of an existence proof. Nor is a particular contact-term convention required when an admissible consistent choice would suffice. The unresolved requirements above concern the actual measure, observable bounds and physical identification.

Verification record and source access

The accompanying research notes preserve the exact finite-measure construction, the self-contained Wilson chessboard proof, the fixed-box conditional proof and their independent internal audits. The assembled source was checked for duplicate and missing labels and compiled with the desktop LaTeX compiler. These checks establish document consistency and typesetting; the mathematical evidence is the arguments and their stated hypotheses.

Primary references used to locate the complete-density interface are T. Bałaban, *Convergent renormalization expansions for lattice gauge theories*, Communications in Mathematical Physics **119** (1988), 243–285, doi:10.1007/BF01217741, and *Large field renormalization. II. Localization, exponentiation, and bounds for the R operation*, Communications in Mathematical Physics **122** (1989), 355–392, doi:10.1007/BF01238433. The exact Section 2 induction class in the first paper, and the weak-coupling hypothesis in Theorem 1 of the second, are retained. No unchecked OCR constants or powers are supplied as theorem hypotheses. The first large-field paper's detailed replacement formula has not been newly verified from an accessible original in this pass; its exact audit remains part of any import route.